

A tutorial of reflection principle in combinatorics

By [-firefly-](#), [history](#), 11 months ago, 

Introduction

A classical problem reads:

Count the paths from $(0, 0)$ to (m, n) ($n > m > 0$) that doesn't go through $y = x$ (the path can touch the line). One can only move up or right by 1 in one move.

The problem can also be described as:

Count binary strings that consist of m zeros and n ones ($n \geq m$), such that in every prefix of the string, the number of zeros doesn't exceed the number of ones.

Consider how to count paths from $(0, 0)$ to (m, n) without the intersection restriction. To get to (m, n) , we must move up n times and move right m times. In other words, we need to permute n right and m up, whose quantity is

$$\frac{(m+n)!}{m!n!} = \binom{m+n}{m}$$

To find the answer, we can try to count the paths that go through $y = x$, and this is when the reflection principle is utilized. The principle says:

Given point A, B at the same side of line l , the number of paths from A to B that intersect with l is equal to that of paths from A' to B , where A' is the reflection point through l .

To show why it is correct, we should first note that a path from A' to B must intersect with $y = x$. Say at point P the path first intersect with $y = x$, we can reflect the path from A' to P through l and get a path from A to B , and vice versa. This indicates that the intersecting paths from A to B and the paths from A' and B are one-to-one corresponded, and the quantity of the two are the same.